

## TECHNICAL PROJECT DOCUMENTATION



*Detection and Spatial Prediction of Hoplocerambyx spinicornis (Sal Borer) Infestation  
in Himalayan Sal (Shorea robusta) Forests Using Sentinel-2 Satellite Intelligence,  
Machine Learning, and a Web-Based Geographic Information System*

Prepared by

**Nitanshu Tak**

B.Tech CSE (Cloud Computing & Virtualization Technology), UPES Dehradun

**Chandreyee Dey Roy**

B.Tech, UPES Dehradun

Under the Supervision of

**Prof. Nadeem Yousuf Khan, UPES Dehradun**

Prepared for submission to the HILL Conference, October 2026  
University of Petroleum and Energy Studies, Dehradun, India



## Abstract

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Uttarakhand is currently experiencing a significant ecological emergency: an active outbreak of the Sal Borer beetle, *Hoplocerambyx spinicornis*, is systematically destroying Sal (*Shorea robusta*) forests across the Dehradun Forest Division, including the Thano, Asarori, and Jhahhra ranges. The Uttarakhand Forest Department has identified more than 19,170 Sal trees for emergency felling as a direct consequence of this infestation, and field reports from the Forest Research Institute (FRI) indicate that the unusually intense 2025 monsoon season created favourable breeding conditions that accelerated the spread of the insect across already vulnerable monoculture plantations.

The detection methodology presently employed by forest departments is entirely reactive in nature. Forest guards patrol assigned beats on foot and visually identify trees that have already died or are exhibiting unmistakable signs of advanced infestation, such as bark sloughing, sawdust accumulation at the tree base, and canopy browning. By the time these visual symptoms become apparent, the underlying larval colonisation has typically already progressed for six to ten weeks, during which period the beetle population has had ample opportunity to disperse to neighbouring trees within the same stand. This reactive paradigm, while necessary given the absence of better alternatives, fundamentally limits the capacity of forest management authorities to intervene before damage becomes irreversible at a landscape scale.

This document presents SAL-SHIELD, a fully implemented and operational forest intelligence platform conceived and developed to invert this reactive paradigm into a proactive, spatially-aware early warning system. SAL-SHIELD operates simultaneously at two complementary spatial scales. At the landscape scale, multitemporal Sentinel-2 satellite imagery is processed to compute a novel composite geospatial risk index, the Sal Borer Vulnerability Index (SBVI), which synthesises vegetation health (NDVI), canopy moisture stress (NDMI), burn-ratio-based degradation signals (NBR), and terrain-derived spread-likelihood factors into a single, interpretable risk score for every pixel of the study area. At the tree scale, a custom-trained convolutional neural network enables forest guards equipped with nothing more than an ordinary smartphone to photograph an individual tree and receive an immediate AI-generated health classification, with the resulting observation geotagged and transmitted to a centralised dashboard the moment network connectivity becomes available.

Two independently trained and independently validated machine learning subsystems underlie the platform. The satellite-scale risk classifier, evaluated using five-fold stratified cross-validation on a set of eighty unambiguously labelled pixels drawn from the SBVI output distribution, achieves 88.8 percent classification accuracy using a Random Forest ensemble and 97.5 percent accuracy using a Gradient Boosted Trees (XGBoost) ensemble. The tree-scale image classifier, a custom three-block convolutional architecture trained from scratch on a curated dataset of 691 field and reference images spanning healthy, stressed, and infested tree classes, achieves 92.7 percent validation accuracy. Both results comfortably exceed the 85 percent accuracy threshold established as the minimum viable performance bar for operational deployment.

The complete system has been implemented as a production-grade web application: a Flask-based REST API backend serving real satellite-derived GeoJSON risk geometries and machine learning inference endpoints, paired with a React-based Progressive Web Application frontend that renders an interactive

Leaflet map of risk zones, provides multitemporal vegetation analytics dashboards, and supports fully offline field data capture through browser-resident IndexedDB storage with automatic background synchronisation upon connectivity restoration. The entire infrastructure — satellite data acquisition, machine learning training and inference, application hosting, and content delivery — operates at zero recurring monetary cost, relying exclusively on freely available European Space Agency and National Aeronautics and Space Administration earth observation assets together with free-tier cloud hosting services.

This document provides an exhaustive technical account of every component of the SAL-SHIELD platform: the ecological and operational context motivating the work; the satellite remote sensing methodology and the derivation of the SBVI formula; the complete machine learning pipeline including feature engineering, model selection, training procedures, and validation results for both subsystems; the software architecture of the backend and frontend applications; the offline-first Progressive Web Application design; the deployment and hosting configuration; and a discussion of current limitations alongside a structured roadmap for future enhancement, including planned collaboration with the Forest Research Institute to obtain field-verified ground truth labels that are expected to further elevate model performance.

# 1. Introduction

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## 1.1 The Ecological Crisis in Uttarakhand

The Sal tree, *Shorea robusta*, is among the most ecologically and economically significant tree species of the Indian subcontinent, forming extensive climax forest communities across the Shivalik foothills and the lower elevations of the Himalayan belt. In Uttarakhand specifically, Sal forests constitute a substantial proportion of the reserved and protected forest estate, providing timber, non-timber forest produce, watershed regulation, and habitat for a wide range of associated flora and fauna. The Dehradun Forest Division, encompassing the Thano, Asarori, Jhajhra, and Kalsi forest ranges, contains some of the most extensive and historically well-managed Sal stands in the state.

Beginning in the period preceding this study and continuing through the present, these forests have come under sustained attack from *Hoplocerambyx spinicornis*, commonly known as the Sal Heartwood Borer. This cerambycid beetle species is a known periodic pest of Sal forests across South Asia, with documented outbreak cycles in the historical silvicultural record. The adult beetle oviposits at the base of the tree or within bark crevices, and the resulting larvae bore into the cambium and heartwood, where they feed and develop over an extended larval period. This larval activity directly severs the vascular cambium responsible for nutrient and water transport between root and crown, and in advanced infestations, the structural integrity of the trunk itself is compromised, rendering affected trees liable to wind-throw even before biological death occurs.

The current outbreak has been attributed by Forest Research Institute field assessments to a confluence of contributing factors. The 2025 monsoon season delivered rainfall substantially above the decadal average across the Dehradun region, and the resulting elevated soil and atmospheric moisture levels are understood to favour beetle breeding success and larval survival rates. Compounding this climatic driver, much of the affected forest estate consists of even-aged monoculture Sal plantations established under historical silvicultural regimes that prioritised single-species timber production. Such monocultures lack the structural and species heterogeneity that, in natural mixed forests, acts as a partial barrier to the unimpeded lateral spread of host-specific pest organisms. The consequence has been rapid radial expansion of the infestation front from initial outbreak foci into adjacent, previously unaffected stands.

The magnitude of the resulting damage is considerable. Departmental felling orders issued in response to confirmed infestation have identified in excess of 19,170 individual Sal trees within the Dehradun Forest Division alone as requiring emergency removal, both to salvage merchantable timber value before complete wood degradation occurs and to remove infestation reservoirs that would otherwise continue to seed further spread. This figure, while substantial, almost certainly understates the true extent of subclinical infestation present within the landscape at any given time, since the detection methodology underlying these felling orders is itself constrained by the limitations discussed in the following section.

## 1.2 Limitations of Current Detection Methodology

Forest health monitoring within the affected divisions presently relies on traditional ground-based surveillance conducted by forest guards and beat-level field staff. This methodology, while institutionally well-established and operationally straightforward, exhibits three structural limitations that are

particularly consequential in the context of a fast-moving biotic disturbance such as the present Sal Borer outbreak.

- ❖ **Detection lag:** visual symptoms of Sal Borer infestation, including canopy yellowing, branch dieback, bark sloughing, and the presence of frass or sawdust at the tree base, typically do not become unambiguously identifiable to a ground observer until the underlying larval colonisation has been active for a period of approximately six to ten weeks. During this latent period, the infestation continues to progress and, critically, the beetle population retains the capacity to disperse to neighbouring host trees, meaning that by the time a single tree is confirmed and reported, the true affected area may already encompass a substantially larger cluster of trees in earlier, undetected stages of colonisation.
- ❖ **Spatial coverage constraints:** ground patrol coverage is necessarily limited by the number of available field personnel relative to the area of forest under their charge. Systematic, exhaustive, tree-by-tree visual inspection of an entire forest division on a frequent recurring basis is operationally infeasible given typical staffing ratios, meaning that large areas of forest may go uninspected for extended intervals, during which an active infestation front can progress substantially.
- ❖ **Absence of predictive or anticipatory capability:** the existing methodology is intrinsically reactive. It can confirm the presence of infestation once visual symptoms manifest, but it provides no mechanism for anticipating which currently asymptomatic areas of forest are at elevated risk of becoming infested in the near future, and consequently offers no basis for the proactive, anticipatory deployment of limited field resources toward areas of greatest emerging concern.

These limitations collectively motivate the central design objective of the work described in this document: the development of a detection and monitoring capability that operates earlier in the infestation timeline than visual ground symptoms permit, that achieves comprehensive spatial coverage independent of field personnel availability through the use of satellite remote sensing, and that provides explicitly predictive, risk-ranked spatial output capable of directing field resources toward areas of elevated concern before, rather than after, visible tree mortality occurs.

### 1.3 Project Objectives

The SAL-SHIELD project was undertaken with the following specific objectives, formulated in consultation with and under the supervision of Prof. Nadeem Yousuf Khan at UPES Dehradun, in direct response to the operational detection gap described above.

1. To develop an original, ecologically grounded composite spectral index, herein termed the Sal Borer Vulnerability Index (SBVI), capable of synthesising multiple satellite-derived vegetation health indicators into a single interpretable risk score correlated with Sal Borer infestation likelihood.
2. To acquire, process, and analyse multitemporal Sentinel-2 satellite imagery across the affected study area, spanning a pre-outbreak baseline period through the current monitoring date, in order to characterise the spectral and temporal signature of the ongoing infestation.

3. To train and rigorously validate machine learning classifiers capable of translating SBVI and associated satellite-derived features into discrete, actionable risk classifications at the level of individual forest pixels, achieving a minimum classification accuracy of 85 percent.
4. To train and validate a complementary image-based machine learning classifier capable of assessing the health status of an individual tree directly from a smartphone photograph, likewise targeting a minimum accuracy of 85 percent.
5. To design and implement a complete, deployable web-based Geographic Information System application that renders the satellite-derived risk classifications as an interactive map, supplemented by temporal analytics, for use by forest department decision-makers.
6. To design and implement a complementary field data collection tool, deployable on ordinary forest guard smartphones without requiring application store distribution, that supports fully offline operation in connectivity-denied forest environments and automatic data synchronisation upon connectivity restoration.
7. To ensure that the entire system, from satellite data acquisition through application hosting, can be operated and replicated at zero recurring monetary cost, in order to maximise the feasibility of operational adoption by resource-constrained forest department authorities.

## 1.4 Document Structure

The remainder of this document is organised as follows. Section 2 provides the satellite remote sensing methodology underlying the platform, detailing the data sources, spectral indices, and temporal sampling strategy employed. Section 3 presents the derivation, formulation, and validation of the Sal Borer Vulnerability Index. Section 4 describes the machine learning pipeline in full technical detail, covering both the satellite-scale risk classification subsystem and the image-scale tree health classification subsystem. Section 5 describes the backend application architecture. Section 6 describes the frontend web application and its constituent modules. Section 7 describes the Progressive Web Application and offline-first architecture supporting field data collection. Section 8 describes the deployment and hosting configuration. Section 9 presents testing methodology and results. Section 10 discusses limitations of the current implementation. Section 11 presents a structured roadmap for future work. Section 12 concludes the document, and is followed by a statement of individual contributions, references, and technical appendices.

## 2. Satellite Remote Sensing Methodology

The satellite remote sensing component of SAL-SHIELD was conceived and executed by Chandreyee Dey Roy, drawing on freely available Earth observation assets to characterise vegetation health and stress across the Dehradun Forest Division study area over a multitemporal sequence spanning the pre-outbreak baseline period through the present monitoring date. This section documents the data sources, processing methodology, spectral indices, and study area specification underlying this work.

### 2.1 Study Area Specification

The study area was defined to encompass the principal Sal forest tracts of the Dehradun Forest Division known to be affected by the current Sal Borer outbreak, including the Thano, Asarori, Jhajhra, and Kalsi forest ranges. The geographic bounding extent of the study area is specified as follows.

| Parameter                   | Value                                    |
|-----------------------------|------------------------------------------|
| Western boundary            | 77.85° E                                 |
| Eastern boundary            | 78.15° E                                 |
| Southern boundary           | 30.20° N                                 |
| Northern boundary           | 30.45° N                                 |
| Approximate area            | ~900 km <sup>2</sup>                     |
| Native pixel resolution     | 10 metres (Sentinel-2 visible/NIR bands) |
| Raster grid dimensions      | 3,340 × 2,785 pixels                     |
| Coordinate reference system | EPSG:4326 (WGS 84 geographic)            |

This bounding region was selected to provide comprehensive coverage of the documented infestation zones while remaining computationally tractable for pixel-level processing on standard workstation hardware, and to align with the administrative boundaries of the Dehradun Forest Division for ease of correlation with departmental felling order records.

### 2.2 Satellite Data Source: Sentinel-2

The Sentinel-2 mission, operated by the European Space Agency under the Copernicus Earth observation programme, was selected as the primary data source for this work on the basis of several characteristics that make it particularly well suited to vegetation health monitoring at the scale required. Sentinel-2 carries a Multispectral Instrument (MSI) sensor providing thirteen spectral bands spanning the visible, near-infrared, and shortwave infrared regions, at native spatial resolutions of 10, 20, or 60 metres depending on the specific band. The constellation, comprising two satellites in sun-synchronous orbit, achieves a combined revisit interval of approximately five days at the latitude of the study area, providing sufficient temporal density to capture seasonal and event-driven vegetation change. Critically for the purposes of this project, the complete Sentinel-2 archive, dating back to mid-2015, is made available by the European Space Agency at no cost under an open data licence, and is additionally accessible through



the Google Earth Engine cloud computing platform, which likewise offers free-tier access for research and educational use.

Imagery was acquired at Level-2A processing, corresponding to bottom-of-atmosphere surface reflectance with cloud masking and atmospheric correction already applied by the European Space Agency processing pipeline, thereby eliminating the need for independent atmospheric correction and substantially simplifying the downstream index computation workflow.

2.3 Temporal Sampling Strategy

In order to characterise the progression of the infestation over time, four discrete temporal scenes were selected and processed, each representing a distinct phase in the documented outbreak timeline.

| Scene Label | Acquisition Period | Significance                                                                                                               |
|-------------|--------------------|----------------------------------------------------------------------------------------------------------------------------|
| Baseline    | October 2023       | Pre-outbreak reference state, representing forest condition prior to the documented onset of the current infestation cycle |
| Pre-Stress  | March 2024         | Early monitoring point capturing initial, subclinical vegetation stress signals preceding widespread visible symptoms      |
| Peak Damage | October 2024       | Scene corresponding to the period of most severe documented canopy damage and departmental felling order activity          |
| Current     | June 2025          | Most recent available scene at the time of analysis, representing the present monitoring state of the forest               |

This four-point temporal design was selected in preference to a simple two-point (before/after) comparison because it permits characterisation of the trajectory of vegetation decline, rather than merely its net magnitude, and because it provides the additional pre-stress sampling point required to identify spectral signals that precede, and may therefore serve as early predictors of, the subsequently observed peak damage state.

2.4 Spectral Indices Computed

From the surface reflectance bands provided by each Sentinel-2 scene, four derived spectral indices were computed, each capturing a distinct aspect of vegetation condition relevant to bark beetle infestation detection. Each index was computed independently for all four temporal scenes, yielding a multitemporal stack of raster layers subsequently used both for direct interpretation and as input features to the machine learning pipeline described in Section 4.

2.4.1 Normalised Difference Vegetation Index (NDVI)

The Normalised Difference Vegetation Index is among the most widely used and well-validated indicators of vegetation photosynthetic activity and canopy density, computed from the near-infrared and red reflectance bands according to the following formula.

NDVI = (NIR - Red) / (NIR + Red)

Healthy, dense Sal canopy characteristically exhibits NDVI values in the range of approximately 0.65 to 0.80, reflecting high chlorophyll content and vigorous photosynthetic activity. As infestation progresses and the larval activity disrupts vascular transport, the affected canopy exhibits progressive chlorosis and eventual necrosis, with NDVI values in severely infested stands declining to the range of 0.25 to 0.45. The whole-study-area mean NDVI was observed to decline from 0.647 at the October 2023 baseline to 0.500 at the June 2025 current monitoring point, a relative decline of approximately 22.7 percent over the twenty-month observation window.

#### 2.4.2 Normalised Difference Moisture Index (NDMI)

The Normalised Difference Moisture Index, computed from the near-infrared and shortwave-infrared reflectance bands, is sensitive to canopy water content and is understood to respond to physiological water stress in vegetation at an earlier stage than the broader photosynthetic decline captured by NDVI.

$$\text{NDMI} = (\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR})$$

Because *Hoplocerambyx spinicornis* larval activity directly compromises the vascular cambium responsible for water transport from root to crown, canopy moisture stress represents a mechanistically direct and early consequence of infestation, preceding the broader chlorophyll degradation that subsequently manifests as declining NDVI. This mechanistic relationship is corroborated by the subsequent machine learning feature importance analysis presented in Section 4, in which NDMI-derived features consistently rank as the most predictive inputs after the SBVI score itself, providing empirical support for the use of NDMI as an early-warning indicator. The whole-study-area mean NDMI was observed to decline from 0.197 at baseline to 0.064 at the current monitoring point.

#### 2.4.3 Normalised Burn Ratio (NBR)

The Normalised Burn Ratio, originally developed for the assessment of post-fire vegetation damage, is computed from the near-infrared and a second, longer-wavelength shortwave-infrared band.

$$\text{NBR} = (\text{NIR} - \text{SWIR2}) / (\text{NIR} + \text{SWIR2})$$

Although developed for fire damage assessment, the NBR index responds generally to severe vegetation degradation and structural canopy loss, irrespective of the underlying causal mechanism, and severely beetle-damaged or dead Sal canopy was found to exhibit a spectral signature broadly comparable to fire-damaged vegetation in this index. NBR was computed for the baseline, peak damage, and current scenes, and was incorporated as a weighted component of the SBVI formula described in Section 3.

#### 2.4.4 Terrain Slope

In addition to the three spectral indices described above, a terrain slope layer was derived from the Shuttle Radar Topography Mission (SRTM) thirty-metre digital elevation model, made freely available by the National Aeronautics and Space Administration. Terrain slope was incorporated into the SBVI formula as a proxy for spread-likelihood, on the basis that beetle dispersal and microclimatic conditions favourable to breeding success are understood to exhibit some dependence on local topographic aspect and gradient.

### 2.5 Data Processing Pipeline

Satellite scene acquisition and initial index computation were performed using the Google Earth Engine cloud computing platform, which provides direct programmatic access to the complete Sentinel-2 archive

together with sufficient computational resources to perform band arithmetic across the full study area extent without requiring local download of the underlying multi-gigabyte source imagery. Following index computation, derived raster outputs were exported in GeoTIFF format and subsequently processed locally using QGIS and the Rasterio Python library for cloud masking refinement, temporal differencing, and the composite SBVI computation described in the following section.

The complete set of processed raster outputs delivered for downstream machine learning use comprised thirteen distinct GeoTIFF layers: NDVI and NDMI at each of the four temporal scenes (eight layers), NBR at the baseline, peak damage, and current scenes (three layers), a single-epoch NDVI difference layer and a single-epoch NDMI difference layer computed between the October 2024 and October 2023 scenes (two layers), and the terrain slope layer derived from the SRTM digital elevation model.

### 3. The Sal Borer Vulnerability Index (SBVI)

The Sal Borer Vulnerability Index represents the central original methodological contribution of this work. At the time of project commencement, no composite spectral index specifically calibrated for the detection of *Hoplocerambyx spinicornis* infestation in Himalayan Sal forest ecosystems had been identified in the available literature, despite the existence of generic vegetation stress indices applied in other forest health contexts. SBVI was developed to address this gap, synthesising the four spectral and terrain-derived components described in Section 2 into a single, weighted composite score interpretable as a continuous measure of infestation vulnerability at the level of an individual satellite pixel.

#### 3.1 Formula Derivation and Component Weighting

The SBVI formula was constructed as a weighted linear combination of four stress components, each normalised to a comparable numerical range prior to combination, with weights assigned on the basis of the relative ecological significance and mechanistic directness of each component's relationship to active beetle infestation.

```
SBVI = (NDVI_stress × 0.35)
      + (NDMI_stress × 0.35)
      + (NBR_stress  × 0.20)
      + (SAR_stress  × 0.10)
```

The NDVI stress and NDMI stress components were each assigned the highest weighting of 0.35, reflecting the central role of vegetation photosynthetic decline and canopy moisture stress as the two most direct and well-established spectral consequences of vascular disruption caused by larval boring activity. The NBR stress component was assigned an intermediate weighting of 0.20, capturing severe structural canopy degradation as a complementary, partially independent stress signal. The terrain-derived spread-likelihood component, denoted SAR stress in reference to its derivation from synthetic aperture radar-compatible terrain slope processing, was assigned the lowest weighting of 0.10, reflecting its role as a secondary modulating factor on baseline spectral stress rather than a primary direct indicator.

Each component term was computed as a normalised stress fraction, expressing the proportional decline of the relevant index value relative to the October 2023 pre-outbreak baseline measurement at the same pixel location, such that a pixel exhibiting no decline relative to baseline yields a stress value approaching zero for that component, while a pixel exhibiting severe relative decline yields a stress value approaching or exceeding one. The resulting composite SBVI score for a given pixel therefore represents a weighted aggregate measure of overall vegetation health decline relative to the pre-outbreak reference condition, scaled such that higher values indicate greater inferred vulnerability to, or evidence of, active infestation.

#### 3.2 Risk Classification Thresholds

In order to translate the continuous SBVI score into actionable, discrete risk categories suitable for direct operational use by forest department field staff, five risk classes were defined on the basis of the empirical distribution of SBVI values observed across the study area, calibrated such that the upper risk classes correspond approximately to the spatial extent of pixels already associated with confirmed departmental felling order activity.

| SBVI Range  | Risk Class | Recommended Action                                          |
|-------------|------------|-------------------------------------------------------------|
| < 0.45      | Very Low   | Routine periodic monitoring; no immediate action required   |
| 0.45 – 0.58 | Low        | Scheduled periodic field inspection at standard interval    |
| 0.58 – 0.68 | Moderate   | Priority field team dispatch for ground verification        |
| 0.68 – 0.78 | High       | Forest Research Institute scientific assessment recommended |
| > 0.78      | Very High  | Immediate intervention and felling order assessment         |

### 3.3 Empirical Results Across the Study Area

Application of the SBVI formula across the full study area extent, restricted to forested pixels following appropriate land-cover masking, yielded 223 pixels with valid, non-null SBVI values, distributed across the defined risk classes as summarised below.

| Risk Class | Pixel Count | Mean SBVI (class) | Approx. Area |
|------------|-------------|-------------------|--------------|
| Very Low   | 87          | 0.363             | 0.87 ha      |
| Low        | 59          | 0.510             | 0.59 ha      |
| Moderate   | 35          | 0.627             | 0.35 ha      |
| High       | 18          | 0.728             | 0.18 ha      |
| Very High  | 24          | 0.886             | 0.24 ha      |

Across the full set of 223 valid pixels, the SBVI score ranged from a minimum of 0.114 to a maximum of 1.081, with a study-area mean of 0.529, indicating an overall moderate level of vegetation stress consistent with the documented active outbreak status of the region. The forty-two pixels classified into the High and Very High risk categories, representing approximately nineteen percent of the analysed forest area, were found upon subsequent visual cross-referencing to correspond closely with the spatial distribution of compartments named in departmental felling order documentation, providing a measure of external corroboration for the validity of the SBVI methodology despite the absence, at the time of this writing, of formally collected field ground truth labels.

### 3.4 Vectorisation for WebGIS Deployment

The raw SBVI raster output, while suitable for analytical processing, required conversion to vector polygon format for efficient rendering within the web-based Geographic Information System application described in Section 6. This conversion was performed using a contour-based polygon extraction procedure operating independently on each of the five risk classes: for each class, all pixels satisfying the corresponding SBVI threshold range were extracted as a binary mask, and contiguous regions within this mask were merged into multipolygon geometries using a union operation. The resulting five multipolygon features, each annotated with its risk class, mean SBVI score, contributing pixel count, and approximate area, were serialised into a single GeoJSON FeatureCollection document for direct consumption by the Leaflet mapping library within the frontend application.

### 3.5 Methodological Caveat and Path to Validation

It is important to state explicitly, in the interest of scientific transparency, that the risk classifications described in this section, and the machine learning models trained upon them as described in Section 4, are presently derived from SBVI-internal thresholds rather than from independently collected, field-verified ground truth labels of confirmed infested versus confirmed healthy trees. This represents a methodological limitation common to remote-sensing-based pest detection studies conducted in the absence of an established field validation programme, and is discussed further in Section 10. A structured plan for addressing this limitation through collaboration with the Forest Research Institute Dehradun, with the objective of obtaining Global Positioning System-tagged field observations of confirmed infestation status suitable for use as independent validation and training labels, is presented in Section 11.

## 4. Machine Learning Pipeline

Two independent machine learning subsystems were designed, trained, and validated as part of SAL-SHIELD, each addressing a distinct spatial scale and data modality. The first, described in Section 4.1, operates on satellite-derived spectral features at the scale of an individual forest pixel and produces the risk classifications underlying the SBVI map described in Section 3. The second, described in Section 4.2, operates on smartphone-captured photographic imagery of individual trees and produces a direct health classification for use by field guards. This section documents the feature engineering, model architecture, training procedure, and validation methodology and results for each subsystem in turn.

### 4.1 Satellite-Scale Risk Classification Model

#### 4.1.1 Feature Engineering

The thirteen raw raster layers described in Section 2.5 were stacked into a unified per-pixel feature matrix, in which each row corresponds to a single forest pixel and each column corresponds to a distinct satellite-derived measurement. In addition to the raw index values at each temporal scene, eight further engineered features were derived to explicitly capture temporal change and relative decline, which preliminary analysis indicated carried substantial additional discriminative information beyond the raw index values alone.

| Feature Category           | Constituent Features                                                                                   |
|----------------------------|--------------------------------------------------------------------------------------------------------|
| Raw index values           | NDVI and NDMI at baseline, pre-stress, peak damage, and current scenes (eight features)                |
| Raw NBR values             | NBR at baseline, peak damage, and current scenes (three features)                                      |
| SBVI raw score             | Pixel-level SBVI value as computed in Section 3 (one feature)                                          |
| Terrain                    | Slope derived from SRTM digital elevation model (one feature)                                          |
| Single-epoch differences   | NDVI difference and NDMI difference, October 2024 minus October 2023 (two features)                    |
| Engineered decline metrics | NDVI total drop, NDMI total drop, NBR change, NDVI relative ratio, NDMI relative ratio (five features) |
| Engineered cross-terms     | Additional differenced and ratio terms derived during feature exploration (one feature)                |

In total, twenty-one features were retained in the final feature matrix following an iterative feature engineering process in which candidate engineered features were evaluated against cross-validated classification accuracy and retained only where they contributed measurable discriminative improvement.

#### 4.1.2 Label Construction

In the absence of independently collected field ground truth labels, training labels for the satellite-scale classifier were constructed using strict, unambiguous thresholds applied to the SBVI score, deliberately excluding the moderate-risk band where genuine ambiguity in ground status is most likely to exist. Pixels with SBVI below 0.35 were labelled as the Healthy class, pixels with SBVI between 0.60 and 0.68 were labelled as the Stressed class, and pixels with SBVI above 0.78 were labelled as the Infected class. This

strict labelling procedure, while necessarily a pseudo-label approach pending field validation as discussed in Section 3.5, yielded eighty pixels with unambiguous class assignment, comprising thirty Healthy, twenty-six Stressed, and twenty-four Infected examples, judged sufficient for the purposes of an initial proof-of-concept classifier subject to the caveats discussed in Section 10.

#### 4.1.3 Model Selection and Training

Two tree-ensemble algorithms were selected for evaluation on the basis of their established strong performance on tabular, moderate-dimensionality feature data of the kind produced by the satellite feature engineering pipeline, together with their relative robustness to the modest training sample size available.

- ❖ Random Forest: an ensemble of 300 decision trees, each trained on a bootstrap-resampled subset of the training data with random feature subsampling at each split, configured with a maximum tree depth of ten and a minimum of two samples per leaf node, with class weights balanced to compensate for the modest class imbalance present in the eighty-pixel training set.
- ❖ XGBoost (Extreme Gradient Boosting): an ensemble of 200 sequentially trained decision trees, each correcting the residual error of the preceding trees, configured with a maximum tree depth of six, a learning rate of 0.05, and row and column subsampling rates of 0.8 to mitigate overfitting on the limited training sample.

Prior to training, all twenty-one input features were standardised using a StandardScaler transformation, rescaling each feature to zero mean and unit variance, in order to ensure comparable numerical scale across features derived from heterogeneous physical units and value ranges.

#### 4.1.4 Validation Methodology and Results

Given the modest training sample size of eighty pixels, a single fixed train-test split was judged insufficient to provide a stable, reliable estimate of generalisation performance, since the resulting accuracy estimate would be subject to substantial variance depending on the particular random partition selected. Five-fold stratified cross-validation was therefore employed in preference, in which the eighty-pixel dataset is partitioned into five class-stratified folds, with each fold serving once as a held-out validation set while the remaining four folds are used for training, and the resulting five accuracy estimates averaged to yield a more stable overall performance estimate together with an associated standard deviation.

| Model                             | Mean CV Accuracy | Standard Deviation | Validation Strategy                |
|-----------------------------------|------------------|--------------------|------------------------------------|
| Random Forest                     | 88.8%            | ± 4.7%             | 5-fold stratified cross-validation |
| XGBoost                           | 97.5%            | ± 3.1%             | 5-fold stratified cross-validation |
| Ensemble (averaged probabilities) | 93.1%            | —                  | Derived from above                 |

Both individual models substantially exceed the 85 percent accuracy threshold established as the minimum acceptable performance bar for this work, with XGBoost in particular achieving a notably strong 97.5 percent mean cross-validated accuracy. It should be noted that, owing to the high-capacity nature of both ensemble algorithms relative to the modest training sample size, both models additionally achieve



100 percent accuracy when evaluated on their own training data; this training accuracy figure is reported here for completeness but should not be interpreted as a generalisation performance estimate, for which the cross-validated figures above are the appropriate reference.

#### 4.1.5 Feature Importance Analysis

Examination of the Random Forest feature importance scores, computed as the mean decrease in node impurity attributable to each feature across all trees in the ensemble, provides insight into which satellite-derived signals carry the greatest discriminative power for infestation risk classification.

| Rank | Feature                                            | Importance Score |
|------|----------------------------------------------------|------------------|
| 1    | sbvi_raw (pixel SBVI score)                        | 0.391            |
| 2    | ndmi_diff (single-epoch NDMI difference)           | 0.073            |
| 3    | ndmi_diff_e (engineered NDMI difference term)      | 0.066            |
| 4    | nbr_change (NBR baseline-to-current change)        | 0.055            |
| 5    | ndmi_total_drop (NDMI baseline-to-current decline) | 0.049            |

The dominance of the SBVI raw score itself as the leading predictor is expected, given that the training labels were directly constructed from SBVI thresholds; this feature’s importance is therefore most appropriately interpreted as confirming the internal consistency of the labelling procedure rather than as an independent validation finding. Of substantially greater scientific interest is the consistent appearance of NDMI-derived features in the second, third, and fifth importance ranks, collectively suggesting that canopy moisture stress signals carry the strongest independent discriminative signal among the available satellite features after the SBVI score itself. This finding is consistent with, and provides quantitative machine-learning-derived corroboration for, the ecological hypothesis articulated in Section 2.4.2: that canopy moisture stress represents a mechanistically direct and comparatively early consequence of the vascular disruption caused by Sal Borer larval activity, and may therefore hold particular promise as an early-warning indicator in future refinements of the SBVI methodology.

## 4.2 Image-Scale Tree Health Classification Model

### 4.2.1 Dataset Construction

Training data for the image-based tree health classifier was assembled from a combination of online reference image repositories, given the practical constraints on collecting a large volume of field-photographed Sal Borer infestation imagery within the project timeline. Three classes were defined: Healthy, Stressed, and Infected.

| Class    | Final Count (post-cleanup) | Primary Sources                                                                                                                        |
|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Healthy  | 211                        | iNaturalist <i>Shorea robusta</i> observation records; Wikimedia Commons Forest and Forestry categories                                |
| Stressed | 274                        | PlantDoc disease dataset — leaf blight, scab, and spot disease categories selected for visual analogy to early-stage vegetative stress |

|          |     |                                                                                                             |
|----------|-----|-------------------------------------------------------------------------------------------------------------|
| Infected | 206 | PlantDoc severe disease categories; Wikimedia Commons Bark_beetle_damage and Coarse_woody_debris categories |
|----------|-----|-------------------------------------------------------------------------------------------------------------|

Following initial assembly, the combined dataset was subjected to an automated integrity validation pass, in which every image file was opened and verified using the Python Imaging Library to confirm that it represented a valid, decodable image rather than a corrupted download, an HTML error page incorrectly saved with an image file extension, or another form of invalid data. This cleaning process identified and removed 202 corrupted or invalid files from the initially assembled dataset, after which supplementary images were sourced to restore adequate per-class sample counts, yielding the final counts of 211, 274, and 206 images for the Healthy, Stressed, and Infected classes respectively, for a total of 691 clean training images.

#### 4.2.2 Model Architecture

A custom convolutional neural network architecture was designed and trained from scratch, in preference to transfer learning from a large pretrained network such as MobileNetV2, on the basis of computational resource constraints encountered during development; specifically, the memory footprint required to load pretrained ImageNet weights for a network such as MobileNetV2, combined with the TensorFlow graph construction and execution overhead, was found to exceed the available memory on the development workstation used for model training. The resulting compact, purpose-built architecture, while smaller in parameter count than a typical pretrained backbone, was found to achieve strong validation performance on the available dataset while remaining trainable within the available memory budget.

| Layer Block           | Configuration                                                                |
|-----------------------|------------------------------------------------------------------------------|
| Input                 | 96 × 96 × 3 (RGB)                                                            |
| Convolutional Block 1 | Conv2D(32) → BatchNorm → Conv2D(32) → MaxPool → Dropout(0.25)                |
| Convolutional Block 2 | Conv2D(64) → BatchNorm → Conv2D(64) → MaxPool → Dropout(0.25)                |
| Convolutional Block 3 | Conv2D(128) → BatchNorm → MaxPool → Dropout(0.3)                             |
| Classification Head   | GlobalAveragePooling2D → Dense(256, ReLU) → Dropout(0.5) → Dense(3, Softmax) |

The reduced 96-by-96-pixel input resolution, in combination with a small training batch size of four images per gradient step, was likewise selected specifically to minimise peak memory consumption during training, in contrast to the 224-by-224-pixel input resolution and larger batch sizes more typical of image classification pipelines built on standard pretrained backbones.

#### 4.2.3 Training Procedure

The 691-image dataset was partitioned into a training subset of 554 images and a validation subset of 137 images, corresponding to an 80/20 split stratified by class. Data augmentation was applied to the training subset only, comprising random rotation of up to thirty degrees, horizontal flipping, zoom of up to twenty percent, brightness adjustment within a 0.7 to 1.3 multiplicative range, and small random width and height shifts, with the objective of improving model robustness to the variation in photographic angle,

lighting condition, and framing expected to be encountered in genuine field-captured smartphone imagery.

Training proceeded over twenty-five epochs using the Adam optimiser with an initial learning rate of 0.001, with early stopping configured to halt training and restore the best-observed weights should validation accuracy fail to improve over six consecutive epochs, and with a learning rate reduction callback configured to halve the learning rate should validation loss plateau over three consecutive epochs. Model checkpointing was configured to retain only the weights corresponding to the epoch achieving the highest observed validation accuracy throughout training.

#### 4.2.4 Validation Results

| Metric                    | Value                           |
|---------------------------|---------------------------------|
| Final validation accuracy | 92.7%                           |
| Training samples          | 554                             |
| Validation samples        | 137                             |
| Number of classes         | 3 (Healthy, Stressed, Infected) |
| Input resolution          | 96 × 96 × 3                     |
| Total training epochs run | 25 (with early stopping)        |

The achieved validation accuracy of 92.7 percent substantially exceeds the 85 percent minimum performance threshold established for this work, and was achieved despite the comparatively compact architecture and reduced input resolution necessitated by hardware memory constraints, suggesting that the visual distinction between the three target classes within the assembled dataset is sufficiently pronounced to be reliably captured even by a relatively lightweight convolutional model. As discussed further in Section 10, it should nonetheless be noted that the training images, while visually representative of the target classes, were predominantly sourced from reference repositories rather than from field photographs of confirmed Sal Borer infestation specifically, and ongoing refinement using genuine field-captured imagery, as outlined in Section 11, is expected to further improve real-world deployment accuracy.

## 5. Backend Application Architecture

---

The SAL-SHIELD backend was implemented as a Flask-based REST API service, selected for its lightweight footprint, minimal configuration overhead, and straightforward integration with the Python-based machine learning libraries used for model inference. This section describes the backend's data persistence layer, machine learning model loading procedure, and the complete set of exposed application programming interface endpoints.

### 5.1 Technology Selection

Flask was selected over alternative Python web frameworks such as Django on the basis that the backend's responsibilities are limited to a focused set of stateless or lightly-stateful API endpoints rather than requiring the fuller content-management and administrative tooling that heavier frameworks provide, and that Flask's minimal abstraction layer simplifies direct integration with the scikit-learn, XGBoost, and TensorFlow inference calls required by the machine learning subsystems described in Section 4.

### 5.2 Data Persistence

Field report data submitted by forest guards through the frontend application, as described in Section 6.2, is persisted using SQLite, a lightweight, file-based relational database engine requiring no separate database server process. This choice was made deliberately to minimise deployment complexity and operational overhead, consistent with the project's zero-cost infrastructure objective; SQLite is fully adequate for the anticipated query volume and concurrency requirements of a field reporting system of this scale, while eliminating the need for a managed database service that would introduce additional hosting cost.

The `field_reports` table schema captures the health status classification, model confidence score, geographic coordinates, severity assessment, free-text field notes, submitting guard identifier, associated image file path, forest zone identifier, synchronisation status flag, and submission timestamp for each individual field report. The database is automatically initialised with five representative demonstration records on first application startup, ensuring that the application dashboard presents meaningful content even prior to the accumulation of genuine field submissions.

### 5.3 Machine Learning Model Integration

Both machine learning subsystems described in Section 4 are loaded into the Flask application process at startup and held resident in memory for the duration of the application's running lifetime, in order to minimise per-request inference latency. The Random Forest model, XGBoost model, and associated feature scaler are deserialised from disk using the Python pickle serialisation format, while the convolutional neural network image classifier is loaded using the standard Keras HDF5 model format. Should the image classifier model file be unavailable, for example prior to completion of the training procedure described in Section 4.2, the application gracefully falls back to a deterministic placeholder prediction function, ensuring that the broader application remains functional for demonstration and development purposes even in the absence of a fully trained image model.

### 5.4 Application Programming Interface Reference

| Method | Endpoint             | Description                                                                                                                                                              |
|--------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GET    | /api/health          | Returns overall system health status, including confirmation of machine learning model load status, GeoJSON data file availability, and database file presence           |
| GET    | /api/risk-zones      | Returns the real, satellite-derived SBVI risk zone geometries as a GeoJSON FeatureCollection, for direct rendering by the frontend mapping component                     |
| GET    | /api/stats           | Returns aggregate summary statistics for dashboard display, including field report counts by status category and current model accuracy figures                          |
| GET    | /api/model-info      | Returns detailed machine learning model metadata, including per-model accuracy figures, feature lists, and the SBVI formula specification                                |
| GET    | /api/ndvi-timeseries | Returns the four-point multitemporal NDVI and NDMI time series data described in Section 2.3, for rendering by the frontend analytics charts                             |
| GET    | /api/reports         | Returns the full list of submitted field reports, with optional filtering by health status category                                                                      |
| POST   | /api/predict/image   | Accepts an uploaded photographic image and returns the tree health classification produced by the convolutional neural network described in Section 4.2                  |
| POST   | /api/field-report    | Accepts and persists a complete field report submission, including health status, geographic coordinates, severity, notes, and an optional associated photographic image |

All endpoints are configured with permissive cross-origin resource sharing headers to support direct invocation from the frontend application regardless of the specific domain or port from which it is served, accommodating both local development and production deployment configurations as described in Section 8.

## 6. Frontend Web Application

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The SAL-SHIELD frontend was implemented as a single-page React application, built using the Vite tooling chain, and structured around five principal functional modules accessible through a persistent navigation interface. This section describes the overall application shell and each constituent module in turn.

### 6.1 Technology Selection and Application Shell

React 18 was selected as the frontend framework on the basis of its mature ecosystem of mapping and charting libraries directly relevant to this project's requirements, specifically React-Leaflet for interactive Geographic Information System map rendering and Recharts for the temporal analytics visualisations described in Section 6.3. Vite was selected as the build tool in preference to alternatives such as Create React App on the basis of its substantially faster development server startup and hot-module-reload performance, and its first-class support for Progressive Web Application configuration through the vite-plugin-pwa plugin, discussed further in Section 7.

The application shell comprises a persistent top navigation bar displaying the application identity, live or offline connectivity status, and pending offline-report synchronisation count; a desktop sidebar providing primary navigation between the five functional modules together with a live-updating summary of current risk zones; and, on narrower mobile viewports, a bottom navigation bar providing equivalent primary navigation in a layout appropriate to touch-based phone interaction.

### 6.2 Dashboard Module

The Dashboard module serves as the primary landing view and presents the core Geographic Information System risk map. The map component, built using React-Leaflet atop a dark-themed CartoDB basemap tile layer, renders the real SBVI risk zone GeoJSON geometries retrieved from the backend, described in Section 3.4, with each of the five risk classes rendered in a distinct colour and individually toggleable through a row of filter controls positioned above the map. Selecting an individual risk zone polygon opens a detail panel presenting the zone's mean SBVI score, contributing pixel count, approximate area, and the underlying SBVI formula. A supplementary panel beneath the map renders a compact multitemporal area chart of the real NDVI and NDMI time series data described in Section 2.3, providing immediate visual context for the vegetation decline trend without requiring navigation to the dedicated Analytics module. An adjacent alert panel surfaces zones that have crossed the High or Very High risk thresholds, intended to draw immediate attention to the areas of greatest current concern.

### 6.3 Field Report Module

The Field Report module constitutes the primary field data collection interface intended for use by forest guard personnel. Upon opening this module, the application immediately attempts to acquire the device's current Global Positioning System location through the browser Geolocation API, with a custom hook implementing graceful fallback behaviour: where genuine device location cannot be obtained, for example because the page is being served over an insecure, non-HTTPS connection on a local development network on which browser security policy restricts Geolocation API access, the application falls back to the geographic centroid of the defined study area, with a clear visual indicator informing the user that a fallback location is in use rather than a genuine device-reported position.

The guard photographs the subject tree using the device camera, invoked directly through the standard HTML file input capture attribute without requiring any custom native camera integration, after which the captured image is submitted to the backend convolutional neural network inference endpoint described in Section 5.4. The returned classification, comprising the predicted health status label and an associated confidence percentage, is displayed immediately, accompanied by a status-appropriate colour coding and a short, plain-language recommended action statement. The guard additionally records a manual severity assessment and free-text field notes before submitting the complete report. Where the device is online at the time of submission, the report, together with the associated photograph, is transmitted directly to the backend for persistence; where the device is offline, the report is instead written to the browser-resident offline storage mechanism described in Section 7.2, ensuring no loss of field-collected data irrespective of connectivity status at the moment of capture.

## 6.4 Analytics Module

The Analytics module presents a more detailed set of temporal and spatial data visualisations than the summary charts embedded within the Dashboard module, including the full four-point NDVI and NDMI multitemporal area chart, a bar chart presenting mean SBVI score disaggregated by risk zone, and a supplementary chart illustrating the correlation between vegetation index decline and the timing of confirmed infestation activity, intended to support more detailed analytical review by forest department decision-makers and to provide illustrative supporting material for the present documentation and associated conference presentation.

## 6.5 Reports Module

The Reports module presents a filterable tabular listing of all field reports submitted through the Field Report module described in Section 6.3, supporting filtering by health status classification, and displaying for each report the predicted status, model confidence, submitting guard, associated forest zone, geographic coordinates, submission timestamp, and synchronisation status, the latter distinguishing between reports that were submitted while online and confirmed received by the backend versus reports that were captured offline and are pending or have completed background synchronisation as described in Section 7.3.

## 6.6 Saved (Offline Manager) Module

The Saved module provides complete visibility into the offline data capture mechanism underlying the field reporting workflow, directly addressing a design requirement that field users be able to understand and trust the system's offline behaviour. The module presents a listing of every report currently held in browser-resident offline storage pending synchronisation, together with explanatory text describing precisely where this data resides, namely the browser's IndexedDB storage mechanism local to the specific device, and how the synchronisation process operates. Controls are provided to manually trigger synchronisation of all pending reports, to export the complete set of pending reports to a downloadable comma-separated-values file as an independent backup mechanism, and to delete individual pending reports where appropriate.

## 6.7 Visual Design System

A custom dark, forest-themed visual design system was developed for the application, employing a colour palette of deep forest greens for background and surface elements, a lighter sage-green accent colour for interactive elements and primary call-to-action controls, and a small set of semantic status colours, specifically green for healthy or low-risk status, amber for moderate-risk or stressed status, and red for high-risk or infected status, applied consistently across the map risk zone rendering, field report classification display, and alert panel components. This consistent semantic colour mapping was designed to support rapid, intuitive visual interpretation of risk and health status across all modules of the application without requiring the user to consult a separate legend or key.



## 7. Progressive Web Application and Offline Architecture

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A core design requirement for SAL-SHIELD's field data collection capability was that it function reliably within forest interior environments in which cellular network connectivity is frequently absent or severely degraded, while avoiding the substantial additional development and distribution overhead associated with building and publishing separate native mobile applications for the Android and iOS platforms through their respective application stores. This requirement was met through the implementation of the field reporting interface as a Progressive Web Application, a web application architecture that supports installation to a device home screen, offline functionality through client-side caching, and background data synchronisation, while remaining accessible directly through a standard mobile web browser without requiring any application store distribution process.

### 7.1 Installation and Distribution

Because the application is implemented as a standard web application augmented with Progressive Web Application capabilities rather than as a platform-specific native application, distribution to forest guard devices requires nothing more than navigating to the application's web address in a mobile browser, at which point supporting browsers, including Chrome on Android and Safari on iOS, present an option to add the application to the device home screen. Once installed in this manner, the application presents and behaves as a standalone application indistinguishable in everyday use from a natively installed application, while requiring no application store review process, no platform-specific build pipeline, and no developer account registration fee.

### 7.2 Offline Data Storage

Client-side offline data persistence is implemented using IndexedDB, a transactional, object-oriented database mechanism built into all modern web browsers and accessible to web applications without requiring any server-side component. When a field report is captured while the device is offline, or when an attempted submission to the backend fails due to a transient connectivity loss, the complete report payload, including the captured photographic image encoded as a data URL, the predicted health classification, geographic coordinates, severity assessment, and free-text notes, is written directly to an IndexedDB object store local to the device. This data persists across application restarts and device reboots until successfully synchronised to the backend or explicitly deleted by the user through the Saved module described in Section 6.6.

### 7.3 Background Synchronisation

The application continuously monitors device network connectivity status through the standard browser online and offline events. Upon detecting a transition from offline to online status, the application automatically iterates through all reports currently held in IndexedDB pending synchronisation, attempting submission of each to the backend field report endpoint described in Section 5.4, and removing successfully submitted reports from local offline storage upon confirmation of successful backend persistence. This automatic synchronisation behaviour requires no manual user intervention, though a manual synchronisation trigger is additionally provided within the Saved module for cases in

which the user wishes to force an immediate synchronisation attempt or to verify synchronisation status directly.

## 7.4 Service Worker and Asset Caching

Application shell assets, including the compiled JavaScript and CSS bundles and static image assets, together with recently accessed map basemap tiles, are cached using a service worker implemented through the Workbox library and configured via the vite-plugin-pwa build plugin. This caching strategy ensures that, once a forest guard has loaded the application at least once while connected to a network, the complete application interface, including a usable, if not perfectly current, rendering of the map basemap for the guard's typical operating area, remains available and fully navigable even in the complete absence of network connectivity, with only the most recent live data updates, such as newly added risk zones, being unavailable until connectivity is restored.

## 8. Deployment and Hosting Infrastructure

A explicit design objective of SAL-SHIELD was that the complete system, encompassing satellite data acquisition, machine learning model training, and application hosting, should be capable of operation at zero recurring monetary cost, in recognition of the resource constraints typically faced by forest department authorities and academic research groups considering operational or pilot adoption of such a system. This section describes the hosting configuration through which this objective is achieved.

### 8.1 Frontend Hosting

The compiled frontend application is deployed to Vercel, a hosting platform offering a generous free tier suitable for the traffic volumes anticipated for a system of this scale, providing automatic, free Hypertext Transfer Protocol Secure (HTTPS) certificate provisioning, global content delivery network distribution, and direct continuous deployment integration with the project's source code repository such that updates pushed to the repository are automatically rebuilt and redeployed without manual intervention. The HTTPS provisioning provided by this hosting arrangement is a practical prerequisite for reliable operation of the Geolocation API used by the Field Report module described in Section 6.3, since contemporary browser security policy restricts access to genuine device location to pages served over a secure connection.

### 8.2 Backend Hosting

The Flask backend application is deployed to Render, a platform-as-a-service provider likewise offering a free tier suitable for the anticipated request volume, configured to serve the application through the Gunicorn production-grade Web Server Gateway Interface server in preference to the Flask development server used during local development. It should be noted, as a known characteristic of the Render free tier specifically, that backend service instances are automatically suspended following a period of inactivity and require a cold-start interval of approximately thirty to sixty seconds to resume serving requests following a period of dormancy; this characteristic, while inconsequential for demonstration and pilot evaluation purposes, would warrant consideration of a paid hosting tier should the system progress to sustained, latency-sensitive operational deployment.

### 8.3 Cost Summary

| Component                           | Provider                                    | Cost                |
|-------------------------------------|---------------------------------------------|---------------------|
| Satellite imagery (Sentinel-2)      | European Space Agency / Google Earth Engine | ₹0 (free open data) |
| Digital elevation model (SRTM)      | NASA                                        | ₹0 (free open data) |
| Frontend hosting                    | Vercel                                      | ₹0 (free tier)      |
| Backend hosting                     | Render                                      | ₹0 (free tier)      |
| Machine learning frameworks         | scikit-learn, XGBoost, TensorFlow           | ₹0 (open source)    |
| Total recurring infrastructure cost | —                                           | ₹0                  |

## 9. Testing Methodology and Results

A multi-layered testing approach was employed to verify both the machine learning model performance and the end-to-end functional correctness of the deployed application. This section summarises the testing methodology and consolidated results across all tested components.

### 9.1 Machine Learning Model Testing

A dedicated evaluation script was developed to systematically verify the performance of both machine learning subsystems against the 85 percent minimum accuracy threshold established for this work. This script independently loads each trained model, executes the appropriate cross-validation or hold-out validation procedure described in Section 4, and produces a consolidated summary report confirming whether each model meets the established performance target. Execution of this evaluation script against the final trained models confirmed that all three model variants, namely the Random Forest and XGBoost satellite classifiers and the convolutional neural network image classifier, meet or substantially exceed the 85 percent accuracy threshold, as summarised in the consolidated results table below.

| Model                                | Result | Threshold Met |
|--------------------------------------|--------|---------------|
| Random Forest (satellite)            | 88.8%  | Yes           |
| XGBoost (satellite)                  | 97.5%  | Yes           |
| Convolutional Neural Network (image) | 92.7%  | Yes           |

### 9.2 Application Programming Interface Testing

Each backend endpoint listed in Section 5.4 was independently tested using direct Hypertext Transfer Protocol requests to confirm correct response structure and status codes, including verification that the health check endpoint correctly reports model load status, that the risk zones endpoint correctly serves the real SBVI-derived GeoJSON geometry described in Section 3.4, and that the image prediction endpoint correctly accepts a multipart image upload and returns a well-formed classification response.

### 9.3 End-to-End Functional Testing

Complete end-to-end functional testing was conducted across the full application, encompassing verification that the Dashboard module correctly renders live risk zone data retrieved from the backend with correct colour-coded risk classification, that the Field Report module correctly captures device geolocation with appropriate fallback behaviour, correctly invokes the camera capture interface, correctly displays the returned health classification with appropriate confidence visualisation, and correctly routes submissions to either direct backend persistence or offline local storage depending on device connectivity status at time of submission. Offline functionality was specifically verified by disabling device network connectivity, completing a field report submission, confirming correct persistence to local IndexedDB storage and corresponding display within the Saved module, subsequently restoring network connectivity, and confirming correct automatic background synchronisation of the previously offline report to the backend.

## 9.4 Cross-Device and Network Testing

The application was tested across multiple access configurations representative of anticipated real-world usage patterns, including direct access on the development workstation, access from a mobile device connected to the same local area network as the development server for preliminary mobile interface verification, and access from a mobile device to the production Vercel-hosted deployment over a standard cellular data connection, the latter configuration being necessary to verify genuine device Geolocation API functionality given the HTTPS requirement discussed in Section 8.1.

## 10. Limitations

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In the interest of scientific rigour and transparency, this section consolidates the principal limitations of the current implementation that should be borne in mind when interpreting the results presented in this document, and which directly motivate the future work programme described in Section 11.

### 10.1 Absence of Independently Verified Ground Truth

As discussed in Section 3.5, the training labels underlying both the satellite-scale and, indirectly through dataset class assignment, the image-scale machine learning models are derived from internally consistent thresholds and curated reference imagery rather than from field observations independently verified against confirmed Sal Borer infestation status by qualified entomological or forestry personnel. While the reported cross-validation and hold-out accuracy figures are methodologically sound estimates of how well each model can reproduce the labelling scheme used to train it, they do not by themselves constitute proof that the labelling scheme itself perfectly captures genuine ground infestation status. The spatial correspondence observed between the model's High and Very High risk classifications and the locations referenced in departmental felling order documentation, noted in Section 3.3, provides a degree of external corroboration, but does not substitute for systematic field validation.

### 10.2 Limited Satellite-Scale Training Sample Size

The eighty-pixel training set underlying the satellite-scale classifier, while sufficient to demonstrate strong cross-validated accuracy within the present labelling scheme, is modest in absolute terms relative to the heterogeneity of forest conditions, canopy density variation, and microclimatic factors present across a study area of this size. Expansion of the training sample, particularly through the incorporation of field-verified labels as described in Section 11.1, is expected to improve both the robustness and the genuine ecological validity of the resulting classifier.

### 10.3 Image Classifier Training Data Provenance

As noted in Section 4.2.4, the image classifier training dataset was assembled predominantly from general-purpose plant disease and forestry reference image repositories rather than from a dedicated corpus of field photographs specifically depicting confirmed Sal Borer infestation symptoms on *Shorea robusta* trees. While the achieved validation accuracy indicates that the three target classes are visually well separated within the assembled dataset, the degree to which this performance will transfer to genuine field-captured imagery of confirmed Sal Borer cases specifically, as opposed to the visually analogous reference imagery used in training, has not yet been independently verified and represents a priority area for the field data collection programme described in Section 11.1.

### 10.4 Backend Hosting Cold-Start Latency

As noted in Section 8.2, the free-tier backend hosting configuration currently employed is subject to automatic service suspension following inactivity, introducing a cold-start latency of approximately thirty to sixty seconds for the first request following a dormant period. This characteristic, while acceptable for demonstration and pilot purposes, would require either a paid hosting tier upgrade or an alternative

architecture, such as a periodic keep-alive request mechanism, prior to sustained operational deployment in a context where consistent low-latency response is required.

## 11. Future Work

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This section presents a structured, phased roadmap for the continued development of SAL-SHIELD beyond the present implementation, directly addressing the limitations identified in Section 10 and outlining a pathway toward sustained operational deployment in collaboration with relevant forestry research and administrative authorities.

### 11.1 Phase 2: Forest Research Institute Ground Truth Collaboration

The highest-priority item in the future work programme is the establishment of a formal collaboration with the Forest Research Institute, Dehradun, with the specific objective of obtaining Global Positioning System-tagged field observations of trees with independently confirmed Sal Borer infestation status, spanning the full range from healthy through early-stage stressed to severely infected condition. Such field-verified labels would directly address the principal limitation identified in Section 10.1, permitting retraining of the satellite-scale classifier against genuine ground truth in place of the current SBVI-threshold-derived pseudo-labels, and would similarly support the collection of a dedicated, field-photographed image dataset directly addressing the limitation identified in Section 10.3. Preliminary outreach toward this collaboration has been initiated, and it is anticipated that even a modest initial set of thirty to forty confirmed field observations per class would be sufficient to meaningfully improve model validity and to support a substantially strengthened results section in any subsequent peer-reviewed publication arising from this work.

### 11.2 Phase 3: Automated Real-Time Satellite Pipeline

The current satellite data processing pipeline involves a substantial degree of manual Geographic Information System processing performed in QGIS, as described in Section 2.5. A natural and valuable extension of this work is the automation of this pipeline through direct programmatic integration with the Google Earth Engine application programming interface, enabling scheduled, fully automated acquisition and processing of newly available Sentinel-2 scenes as they are published, with corresponding automatic recomputation of the SBVI risk map and automatic refresh of the deployed dashboard, eliminating the present requirement for manual intervention to incorporate new satellite observations.

### 11.3 Phase 4: Generalisation to Additional Forest Types and Pest Species

While the present implementation is calibrated specifically for *Hoplocerambyx spinicornis* infestation in *Shorea robusta* forest, the underlying SBVI methodology, machine learning pipeline architecture, and web application platform are designed to be substantially reusable for other forest health monitoring contexts. A valuable direction for future work is the extension and recalibration of the SBVI formula and associated machine learning models for application to other locally significant forest types present in Uttarakhand, such as Chir Pine or Banj Oak forest, and to other bark beetle or wood-boring pest species of regional concern, with the expectation that the core architecture would require primarily a recalibration of index weightings and retraining against locally appropriate ground truth data rather than a fundamental redesign.

### 11.4 Phase 5: Forest Department Operational Integration



The ultimate objective of this work is operational adoption by forest department field and management personnel. Future work toward this objective includes the development of direct integration interfaces with existing Uttarakhand Forest Department information management systems, the establishment of a formal pilot deployment programme in collaboration with departmental field staff in one or more affected forest ranges, the introduction of role-based access control distinguishing between field guard, range officer, and divisional management user roles within the application, and the incorporation of structured user feedback collected during pilot deployment into subsequent interface and workflow refinement.

## 12. Conclusion

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This document has presented SAL-SHIELD, a complete, operational, and rigorously validated artificial intelligence and Geographic Information System platform for the early detection and spatial prediction of *Hoplocerambyx spinicornis* infestation in Himalayan Sal forests, developed in direct response to an ongoing and ecologically significant outbreak affecting the Dehradun Forest Division of Uttarakhand. The platform combines an original, ecologically grounded composite spectral risk index, the Sal Borer Vulnerability Index, derived from multitemporal Sentinel-2 satellite imagery, with two independently trained and rigorously cross-validated machine learning classification subsystems achieving 88.8 percent, 97.5 percent, and 92.7 percent accuracy respectively, integrated into a fully deployed, production-grade web application supporting both strategic landscape-scale risk visualisation for forest department decision-makers and tactical, offline-capable field data collection for forest guard personnel.

The entire system has been demonstrated to operate at zero recurring monetary infrastructure cost, relying exclusively on freely available satellite Earth observation data and free-tier cloud hosting services, a characteristic considered particularly important for the realistic prospects of operational or pilot adoption by resource-constrained forest department authorities. While the present implementation is subject to the methodological limitations discussed candidly in Section 10, principally relating to the present absence of independently field-verified ground truth labels, a clear and actionable pathway toward addressing this limitation through collaboration with the Forest Research Institute Dehradun has been articulated in Section 11, and the underlying SBVI methodology, machine learning pipeline, and software architecture have each been designed with future extensibility and generalisation to additional forest health monitoring contexts explicitly in mind.

It is the hope of the authors that this work contributes not only a technically sound and immediately deployable tool for the specific and pressing challenge of the current Sal Borer outbreak, but also a methodologically transferable template for the application of freely available satellite remote sensing and modern machine learning techniques to forest health monitoring challenges more broadly, within the resource constraints characteristic of forest department authorities across India and other regions facing comparable ecological and operational circumstances.

## 13. Individual Contributions and Acknowledgements

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### 13.1 Nitanshu Tak

B.Tech Computer Science and Engineering (Cloud Computing and Virtualization Technology), University of Petroleum and Energy Studies, Dehradun. Nitanshu Tak was responsible for the overall system architecture and integration of the platform, the design, training, and validation of both machine learning subsystems described in Section 4, the implementation of the Flask backend application and its complete REST API described in Section 5, the implementation of the React frontend web application and its five constituent functional modules described in Section 6, the design and implementation of the Progressive Web Application and offline-first data collection architecture described in Section 7, and the configuration of the deployment and hosting infrastructure described in Section 8.

### 13.2 Chandreyee Dey Roy

B.Tech, University of Petroleum and Energy Studies, Dehradun. Chandreyee Dey Roy was responsible for the satellite remote sensing methodology described in Section 2, including the acquisition and processing of multitemporal Sentinel-2 imagery across the study area, the computation of the NDVI, NDMI, NBR, and terrain slope derived layers, the original derivation and formulation of the Sal Borer Vulnerability Index described in Section 3, and the production of the underlying GeoTIFF raster and vector shapefile data products that form the foundation of the platform's satellite-derived risk classification capability.

### 13.3 Faculty Supervision

This project was conceived and conducted under the supervision of Prof. Nadeem Yousuf Khan, University of Petroleum and Energy Studies, Dehradun, whose guidance shaped the project's formulation in direct response to the ongoing Sal Borer ecological emergency in Uttarakhand and its targeting toward presentation at the HILL Conference, October 2026. The authors gratefully acknowledge Prof. Khan's sustained guidance and support throughout the conception, execution, and documentation of this work.

## 14. References and Data Sources

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The following data sources, software libraries, and frameworks were employed in the development of this work. Where applicable, the freely available, open-access nature of each resource is noted, consistent with the zero-cost infrastructure objective discussed in Section 8.

- ❖ European Space Agency, Copernicus Sentinel-2 Mission. Multispectral surface reflectance imagery, Level-2A processing. Freely available open data.
- ❖ National Aeronautics and Space Administration, Shuttle Radar Topography Mission (SRTM). Thirty-metre digital elevation model. Freely available open data.
- ❖ Google Earth Engine. Cloud-based geospatial analysis platform used for satellite scene acquisition and index computation. Free tier for research and educational use.
- ❖ QGIS Development Team. QGIS Geographic Information System, used for raster and vector processing.
- ❖ Pedregosa, F. et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, used for the Random Forest classifier implementation described in Section 4.1.
- ❖ Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System. Used for the gradient boosted tree classifier implementation described in Section 4.1.
- ❖ Abadi, M. et al. TensorFlow: Large-Scale Machine Learning on Heterogeneous Systems. Used for the convolutional neural network implementation described in Section 4.2.
- ❖ iNaturalist. Community science biodiversity observation platform, used as a source of *Shorea robusta* reference imagery for the Healthy class of the image classifier training dataset described in Section 4.2.1.
- ❖ Kayal, P. et al. PlantDoc: A Dataset for Visual Plant Disease Detection. Used as a source of reference imagery for the Stressed and Infected classes of the image classifier training dataset described in Section 4.2.1.
- ❖ Wikimedia Commons. Open media repository, used as a supplementary source of forest, forestry, and bark beetle damage reference imagery.
- ❖ Uttarakhand Forest Department. Departmental felling order records pertaining to the Dehradun Forest Division, referenced as contextual evidence of documented infestation extent.
- ❖ React, Vite, Leaflet, React-Leaflet, and Recharts open-source JavaScript libraries, used in the implementation of the frontend web application described in Section 6.
- ❖ Flask, Flask-CORS, and Gunicorn open-source Python libraries, used in the implementation of the backend application described in Section 5.

## Appendix A: Repository Structure

The complete SAL-SHIELD source code repository is organised into two principal top-level directories, corresponding to the backend and frontend applications described in Sections 5 and 6 respectively.

```
sal-shield/
├── backend/
│   ├── app.py           Flask API – all endpoints
│   ├── train_cnn.py      CNN image classifier training script
│   ├── train_model.py    Satellite RF/XGB training script
│   ├── test_model.py     Model evaluation script
│   ├── requirements.txt
│   └── models/
│       ├── rf_model.pkl   Random Forest (88.8%)
│       ├── xgb_model.pkl  XGBoost (97.5%)
│       ├── scaler.pkl     Feature scaler
│       ├── model_meta.json Accuracy and feature metadata
│       └── tree_classifier.h5 CNN image model (92.7%)
│   └── data/
│       └── risk_zones.geojson Real SBVI zones from Sentinel-2
├── frontend/
│   └── src/
│       ├── App.jsx       Shell, routing, navigation
│       └── pages/
│           ├── Dashboard.jsx GIS map, alerts, NDVI chart
│           ├── FieldReport.jsx Camera, GPS, AI classification
│           ├── Analytics.jsx  Temporal NDVI/SBVI analysis
│           ├── Reports.jsx    Field reports table
│           └── Saved.jsx      Offline report manager
│       ├── data/forestData.js Study area bounds, real values
│       ├── hooks/           GPS, toast notifications
│       └── utils/api.js      API calls, offline IndexedDB queue
```

## Appendix B: Glossary of Terms

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| Term      | Definition                                                                                               |
|-----------|----------------------------------------------------------------------------------------------------------|
| SBVI      | Sal Borer Vulnerability Index — the composite spectral risk index defined in Section 3                   |
| NDVI      | Normalised Difference Vegetation Index — vegetation photosynthetic activity indicator                    |
| NDMI      | Normalised Difference Moisture Index — canopy moisture stress indicator                                  |
| NBR       | Normalised Burn Ratio — severe vegetation degradation indicator                                          |
| CNN       | Convolutional Neural Network — the image-based tree health classifier described in Section 4.2           |
| PWA       | Progressive Web Application — the offline-capable field reporting architecture described in Section 7    |
| GeoJSON   | A standard open format for encoding geographic vector data, used to serve risk zones to the frontend map |
| IndexedDB | A browser-resident database mechanism used for offline field report storage                              |